

CI Board — PASS 1, PARALLEL. The 26 partitioned across the CIs. (Casey directive 2026-07-02)

STOP converging on one item. Each CI owns a bucket and works ALL of it this pass — no one waits on the d_2 /down-row bottleneck. Goal: coverage — every one of the 26 gets at least one scored derivation attempt this pass. We want to see how many close.

The three steps, per CI, for every item in your bucket: 1. Scan the corpus for the existing best closed form (K-audits, theorems, toys, ledgers). Fill it into the derivation ledger. 2. Fill the ledger — `play/bst_derivation_ledger.py`, one attempt entry per form: {form, value, how-it-derives-from-substrate, provenance}. Never delete a tried form (so we don't repeat). 3. Attempt an independent derivation from the substrate {rank=2, N_c=3, n_C=5, C_2=6, g=7, N_max=137}. Add it as a new attempt. Not fishing — examining the substrate for a forced derivation of the SM value.

Then hand your filled bucket to Keeper, who σ -scores it and adjudicates the mechanism. You do NOT self-certify — the ledger computes MATCH/APPROX/MISS.

The partition (26 items, 3 buckets)

LYRA — the MIXING sector (8): CKM + PMNS angles & phases

- CKM: θ_{12} (Cabibbo, 9/40 — MATCH 0.6σ), θ_{13} ($V_{ub}(1/3)^5$ — MISS), θ_{23} (V_{cb} dual- ρ), δ_{CKM} (Jarlskog)
- PMNS: θ_{12} (5/16 or 3/10), θ_{13} (1/45 — MATCH 0.5σ , banked), θ_{23} (upper octant), δ_{PMNS} (OPEN — no form yet)
- Your lane: rep-theory of mixing. Pin CKM refs (pdgLive HTML — you have web access; Cal #22 transcription discipline). δ_{PMNS} is your one blank — attempt a form.

GRACE — the MASS sector (9): all charged fermion masses

- Leptons: m_e (anchor/gravity route), m_μ ($((24/\pi^2)^6$ — APPROX), m_τ (49.71 — APPROX)
- Up-type: m_u (up-ladder, honest-neg), m_c (scheme-trap, honest-neg), m_t ($v/\sqrt{2}$ — MATCH scheme-aware)
- Down-type: m_d , m_s , m_b (down-row GJ — MISS, repairable via F453 at observed ladder)
- Your lane: masses via strata/geometry. The down-row d_2 is IN here but is ONE of nine — do NOT let it eat the pass. Cover m_u , m_c , m_e first-pass too. (Lyra handed you the KW boundary reference — verify fresh against Loos before banking $k_2/k_1=20$.)

ELIE — the SCALES + GAUGE + θ_{QCD} + NEUTRINOS (9)

- Gauge: α (1/137 flat & Wyler closed form — pin both), α_s (RG-runner $\rightarrow$ terminal-NEG), $\sin^2\theta_W$ (RG-runner $\rightarrow$ terminal-NEG)
- Higgs: v (pure scale $\rightarrow$ terminal-FLOOR), m_H (λ_H form — OPEN, attempt)
- Strong CP: θ_{QCD} (0 — MATCH, exact, DONE)
- Neutrinos: m_{ν_1} , m_{ν_2} , m_{ν_3} (OPEN — absolute scale is FLOOR; the m_3/m_1 π -free falsifier is your form to score)
- Your lane: numerics + scales. Confirm the 3 terminal-NEG + 1 FLOOR fast (honest coverage), pin α 's Wyler form exactly, attempt m_H and the ν structure.

Scoring (Keeper owns, computed — not declared)

Every attempt scored on BOTH: dev% AND $\sigma = |\Delta|/\text{experimental-error}$ (K643). Use scheme-aware errors for quark masses (m_t reads 4.7σ on the tight pole error but 0.7σ scheme-aware — the error must reflect scheme ambiguity or the metric manufactures misses). Verdict: MATCH ($\sigma \leq 2$) / APPROX ($\sigma > 2$, dev < 1%) / MISS. tol flag = dev $\leq 0.1\%$.

Rules for this pass

- Cover your whole bucket before deep-diving any single item. First pass = at least one scored attempt on all 26.
- A GUT/GUT-scale value is a MISS even with a clean mechanism (down-row). Five-Absence first — Elie's flag: watch the $27=E_6$ and $45=SU(5)-45$ coincidences.
- Structure-forcing, not value-reaching. Don't fish a form to hit a number (137 is form-cheap). Record the substrate derivation, not just the value.
- PDG re-pin your bucket's references (value AND scheme-aware error) as step 0 — the scoreboard's reference can't be unverified.
- Fresh reference work over rushed — Grace's KW pin, Elie's CKM pin: do them carefully, cite primary source, don't wing (genus-flip / Cal #22).

Keeper (me) — loop-keeper, not a deriver

Run SOD check first; keep the derivation ledger; σ -score + mechanism-adjudicate each CI's filled bucket; hold the honest tally; never rewrite a CI's derivation, only score it.

Cal — post-landing cold-read. α tier-call reserved for Casey.

Pass-1 target

Every one of the 26 with ≥ 1 scored attempt. Report the honest tally: MATCH / APPROX / MISS / OPEN. Today's seed (11 of 26): 4 MATCH + 4 APPROX + 3 MISS. Pass 1 fills the other 15 and improves the APPROX/MISS toward MATCH. Then Pass 2 sweeps again — improve forms, add attempts — until the board is all-terminal and Casey calls done.

— Keeper, Pass 1 parallel. Three buckets, three steps each (scan $\rightarrow$ fill $\rightarrow$ derive), coverage over depth this pass. Nobody locked on the down-row. Hand filled buckets to Keeper; the ledger computes the verdict.